Volume Control Transfers, Conditioning Does Not: The Locality of Implicit EM in Neural Networks

Abstract

Gradient descent on log-sum-exp objectives produces responsibility-weighted gradients, so standard neural objectives can instantiate generalized EM without explicit latent-variable updates. But this identity is local: it holds at the objective interface, and does not imply that a whole deep network inherits EM's optimization properties. We characterize this boundary. The Hessian of an LSE objective in energy coordinates is uniformly bounded by 1/2; with private component parameters this yields parameter-independent curvature bounded by a data constant. When the same site is composed behind a learned dense map, the curvature contains a conjugated softmax Hessian, $W^\top(\operatorname{diag} p - pp^\top)W$, whose upper bound scales with the learned spectrum. Thus composition forfeits the parameter-independent guarantee; the measurements show that the composed curvature is in fact larger, non-monotone, and parameter-dependent in the studied network.

Experiments in supervised MNIST networks confirm the split. Volume control applied at an intermediate EM site eliminates dead units and redundancy across widths, showing that local collapse failures transfer. But the optimization signatures observed in the single-layer implicit-EM model disappear under ordinary supervised composition. Measuring gradients shows that the cross-entropy path dominates the local EM path by 30–70× and is nearly orthogonal to it. Cutting that path with stop-gradient restores learning-rate insensitivity and convergence-time invariance; increasing the EM weight produces a continuous transition. The result is a local, graded, structural account of implicit EM: volume control repairs failures at EM sites, but EM conditioning survives only when responsibility gradients reach private parameters through simplex-compatible operations.

1 Notation and Conventions

This section fixes the sign convention used throughout the paper. We use *energy* and *distance* interchangeably: lower values indicate better matches. For an input x and K components, d_j denotes the energy of component j. Responsibilities are defined by

$$r = \operatorname{softmax}(-d), \qquad r_j = \frac{\exp(-d_j)}{\sum_k \exp(-d_k)}.$$

Thus $r_j \geq 0$ and $\sum_j r_j = 1$. In this convention, the log-sum-exp objective is the negative log marginal

$$L_{\text{LSE}}(d) = -\log \sum_j \exp(-d_j),$$

and differentiating with respect to a distance gives

$$\frac{\partial L_{\text{LSE}}}{\partial d_j} = r_j.$$

The sign is positive because increasing a distance makes the negative log marginal larger. If one instead writes logits or negative energies, the same identity appears with the opposite sign. All results below use the distance convention above.

For supervised classification, $h \in \mathbb{R}^C$ denotes class logits and

$$p = \mathrm{softmax}(h)$$

denotes the output-layer class posterior. The label-clamped cross-entropy gradient is

$$\frac{\partial L_{\mathrm{CE}}}{\partial h_j} = p_j - \mathbf{1}\{j = y\}.$$

We reserve r for component responsibilities at an intermediate EM site and p for output class probabilities.

The supervised model has two maps of interest. The first map, W_1, sends the input to component energies. Its rows are treated as private component parameters in the clean single-site theory. The second map, W_2, sends component coordinates to class logits. It is the learned dense component-to-class map, or *corridor*, through which the supervised gradient is pulled back. The largest singular value of this corridor is written $\sigma_{\max}(W_2)$.

The NegLogSoftmin calibration used in the experiments is

$$\mathrm{NLS}(d)_j = d_j + \log \sum_k \exp(-d_k).$$

It preserves pairwise distance differences and satisfies

$$\exp(-\,\mathrm{NLS}(d)_j) = r_j.$$

Thus it converts raw distances into calibrated distances whose exponentials are responsibilities.

Volume control consists of an anti-collapse variance penalty L_{var} and an anti-redundancy decorrelation or total-correlation penalty L_{tc}. We write

$$L_{\mathrm{aux}}(d) = L_{\mathrm{LSE}}(d) + L_{\mathrm{var}}(d) + L_{\mathrm{tc}}(d)$$

for the local auxiliary EM objective when all terms are active. Unless otherwise stated, λ denotes the total auxiliary weight in the supervised objective,

$$L_{\mathrm{total}} = L_{\mathrm{CE}} + \lambda L_{\mathrm{aux}}.$$

In the early ablation experiments, $\lambda_{\mathrm{reg}} = 0.001$ unless noted.

Finally, we use the spectral norm of the LSE Hessian in distance coordinates,

$$\|H_d L_{\mathrm{LSE}}\|_2,$$

as a site-level sharpness diagnostic. Since the LSE Hessian is bounded above by $1/2$, we refer to proximity to this bound as the $1/2$-*saturation diagnostic*. Saturation indicates sharp two-way component competition at the site; values far below $1/2$ indicate diffuse responsibilities.

Symbol	Meaning
x	input example
K	number of intermediate components
C	number of output classes
d_j	energy/distance of component j; lower is better
$r = \text{softmax}(-d)$	responsibility vector over components
h	output logits
$p = \text{softmax}(h)$	output class posterior
W_1	input-to-component map; private rows in the clean theory
W_2	component-to-class corridor; shared learned dense map
$\sigma_{\max}(W_2)$	largest singular value of the corridor
$\text{NLS}(d)$	NegLogSoftmin calibrated distances
L_{LSE}	local log-sum-exp objective
L_{var}	variance penalty, anti-collapse term
L_{tc}	decorrelation/total-correlation penalty, anti-redundancy term
L_{aux}	local EM auxiliary objective
λ	weight on L_{aux} in supervised training
H_d	Hessian with respect to the distance coordinates d

Table 1: Notation used throughout the paper.

2 Introduction

2.1 Starting Point

Prior work established that objectives with log-sum-exp structure over distances or energies produce responsibility gradients. In the convention of this paper,

$$\frac{\partial L_{\text{LSE}}}{\partial d_j} = r_j.$$

Responsibilities therefore arise as gradients, and standard neural objectives can instantiate implicit generalized EM at the objective interface.

The decoder-free encoder study showed that this identity is prescriptive in a single-layer unsupervised model. An objective built from LSE and volume control learned structured prototype-like features, and removing components of volume control produced the predicted collapse and redundancy failures. It also exhibited a distinctive optimization signature: SGD was insensitive to learning rate over a wide range, Adam offered little advantage, and convergence time was roughly fixed.

2.2 The Temptation

Two tempting inferences follow from the identity, and both require a boundary. First, one might infer that whole neural networks inherit EM conditioning because responsibility gradients flow through backpropagation. Second, one might infer that stacking LSE or softmax sites creates a distributed EM mechanism in which depth itself acts like an annealing schedule. Both claims require the EM property to survive learned composition.

This paper asks when the implicit-EM optimization property holds and what destroys it.

2.3 Main Claim

Implicit EM is local, graded, and structural. It is local because the property exists at exponentiation-and-normalization sites and does not automatically extend through learned dense maps. It is graded because real networks interpolate between EM-conditioned and ordinary deep-network behavior according to the measured share of EM-structured gradient reaching the parameters. It is structural because the failures split into at-site failures, which local volume control can repair, and structural failures, which require isolation, dominance, or simplex-compatible transport.

The central mechanism is that backpropagation composition is parameter sharing in disguise. Classical EM loses its clean M-step when components no longer own private parameters and the expected complete-data objective stops separating. The same failure reappears inside neural networks when responsibility gradients are pulled through shared learned maps.

This structural claim must be separated from a dominance claim. Sharing through the supervised corridor explains why the cross-entropy contribution reaching W_1 is not EM-conditioned: the corresponding M-step is non-separable, and its curvature depends on the learned spectrum of W_2. But the gradient arriving at W_1 is a sum of this foreign contribution and the local EM contribution. Whichever contribution dominates sets the effective landscape. Thus full connectivity need not destroy conditioning when the EM term dominates; conversely a small EM term can be drowned by an ill-conditioned supervised path. This is why the paper's claim is graded rather than binary.

2.4 Contributions

This paper makes seven contributions. First, it proves the conditioning contrast between private-parameter LSE objectives and composed dense maps. Second, it classifies operations that preserve or destroy the responsibility-gradient invariant. Third, it introduces the at-site versus structural failure taxonomy. Fourth, it shows experimentally that volume control transfers to intermediate supervised layers and repairs dead or redundant units. Fifth, it shows that optimization conditioning does not survive ordinary composition. Sixth, it establishes causally that conditioning is restored by stop-gradient or EM dominance. Seventh, it introduces the 1/2-saturation diagnostic for mixture sharpness.

3 Related Work and Positioning

EM as an optimization object. The expectation-maximization algorithm and the missing-information principle provide the classical reference point for this work [4]. The free-energy view of EM makes the coordinate-ascent structure explicit [10]. EM has also been analyzed as a preconditioned gradient method [13], which is the closest prior framing to our claim that EM conditioning is structural. The uniform 1/2 curvature bound we use follows the multinomial-logistic Hessian bound of Bohning [2] and the Popoviciu variance inequality.

Classical EM failure modes. Classical EM is cleanest when the expected complete-data objective separates over component-private parameters. Shared parameters, non-separable M-steps, singular components, hard assignments, and diffuse responsibilities are known failure modes. This paper identifies their neural counterparts inside the chain rule.

Generative versus discriminative training. The empirical distinction between generative and discriminative training is well known [11]. We do not claim merely that the objectives differ. We identify a mechanism inside one architecture: a curvature dichotomy, measured gradient dominance, near-orthogonality of the CE and EM paths, and causal restoration under stop-gradient.

Loss landscapes and basin structure. The basin analysis uses linear mode connectivity and permutation alignment ideas from the neural-network loss-landscape literature [5, 6]. Alignment is needed because raw interpolation between component models can create artificial barriers from permutation symmetry alone.

Layer-wise objectives. Greedy layer-wise pretraining and deep belief networks showed that useful representations can be learned by protected local objectives [8]. The same design space includes decoupled greedy layer-wise training [1], synthetic gradients and decoupled neural interfaces [9], and Forward-Forward [7]. This paper gives an implicit-EM mechanism for why local objectives can preserve structure that end-to-end backpropagation does not.

Gradient conflict and auxiliary losses. We measure dominance ratios and cosines between a main supervised loss and a local auxiliary EM loss, placing the experiments near the multi-task gradient-interference literature. Gradient surgery methods such as PCGrad [14], adaptive loss weighting such as GradNorm [3], and gradient starvation analyses [12] all study cases where one training signal degrades or overwhelms another. Our goal is different: we do not propose a balancing method. We use the ratio as a causal control variable for a specific structural property, EM conditioning. In our measurements the gradients are not strongly opposed; the local EM signal is drowned by a dominant foreign signal.

Scope. This is not a general theory of deep-network conditioning. The claims are about EM sites and the corridors through which their responsibility gradients travel.

4 Background: Where EM Lives

4.1 Gradient-Responsibility Identity

For an LSE objective over distances,

$$L_{\mathrm{LSE}}(d) = -\log \sum_j \exp(-d_j), \qquad r = \mathrm{softmax}(-d),$$

the gradient with respect to the distance coordinate is

$$\frac{\partial L_{\mathrm{LSE}}}{\partial d_j} = r_j.$$

For cross-entropy classification,

$$L = -h_y + \log \sum_k \exp(h_k), \qquad \frac{\partial L}{\partial h_j} = \mathrm{softmax}(h)_j - \mathbf{1}[j = y].$$

The output softmax is therefore a label-clamped responsibility interface.

4.2 Locality

The identity applies where the loss sees distances or energies. It does not make all earlier hidden layers into mixture models. The output loss geometry imposes EM structure at the output interface, and gradients from that structure flow backward to shape internal representations. But by the time the gradient has passed through intervening Jacobians, it is no longer a posterior over the earlier layer's units.

4.3 Volume Control

The decoder-free encoder study used volume control to prevent the standard degeneracies of mixture models. The LSE term creates competition and responsibility structure. The variance term prevents collapsed or dead components. The decorrelation term prevents redundant components. Together they play the neural role of a log-determinant barrier.

5 Theory: The Conditioning Boundary

5.1 The Simplex Invariant

Lemma 1 (Simplex invariant and curvature bound)**.** *For*

$$L_{\mathrm{LSE}}(d) = -\log \sum_j \exp(-d_j), \qquad r = \mathrm{softmax}(-d),$$

the gradient $\nabla_d L_{\mathrm{LSE}} = r$ lies on the probability simplex, and the Hessian is

$$H_d L_{\mathrm{LSE}} = \mathrm{diag}(r) - rr^\top, \qquad \|H_d L_{\mathrm{LSE}}\|_2 \le \frac{1}{2}.$$

The proof is the standard covariance form of the multinomial-logistic Hessian, together with the Bohning or Popoviciu bound. The important point here is that the bound is uniform in d and independent of parameters.

5.2 Private Parameters Inherit the Bound

Proposition 1 (Conditioning under private energies)**.** *Let each component energy be a private affine map*

$$d_j = w_j^\top x + b_j.$$

Then

$$\|\nabla_\theta L_{\mathrm{LSE}}\|_2 \le \|x\|, \qquad \|H_\theta L_{\mathrm{LSE}}\|_2 \le \frac{1}{2}\|x\|^2.$$

Under batch-mean training,

$$\|H_\theta L_{\mathrm{LSE}}\|_2 \le \frac{1}{2}\mathbb{E}\|x\|^2,$$

a data constant independent of the current parameters.

This proposition explains the single-layer optimizer signature: the admissible learning-rate range does not shrink as parameters move, curvature is not learned or anisotropic, and an adaptive optimizer has little to correct. The constant is also quantitative. For a batch-mean objective with smoothness constant $L \le \frac{1}{2}\mathbb{E}\|x\|^2$, the standard descent lemma gives a stable gradient-descent step size on the order of $2/L$. We therefore report $\mathbb{E}\|x\|^2$ for the dataset and compare the resulting scale to the observed learning-rate tolerance in the stop-gradient sweep.

Proposition 2 (Smooth monotone kernels). *For $d_j = \phi(w_j^\top x + b_j)$ with ϕ monotone and $|\phi'|, |\phi''|$ bounded, the same parameter-independence survives up to constants determined by the kernel bounds.*

Softplus is the main smooth case. ReLU satisfies the curvature bounds almost everywhere but has a separate absorbing-state pathology: once a unit receives zero gradient, it may never recover.

5.3 Composition Forfeits the Uniform Bound

Proposition 3 (Composition forfeits the uniform bound). *Let*

$$y = \mathrm{NLS}(d), \qquad h = W_2 y, \qquad L_{\mathrm{CE}} = \mathrm{CE}(h, c),$$

and $p = \mathrm{softmax}(h)$*. Then the Hessian of L_{CE} with respect to d contains the conjugated softmax Hessian*

$$J^\top W_2^\top (\mathrm{diag}(p) - pp^\top) W_2 J,$$

plus curvature terms from the NLS map. Consequently,

$$\|H_d L_{\mathrm{CE}}\|_2 = O(\sigma_{\max}(W_2)^2).$$

This proposition is a guarantee-forfeiture statement. It shows that the parameter-independent 1/2 bound of the local LSE site is no longer available; it does not, by itself, prove that the curvature must be large. The empirical curvature measurements in Section 9 carry that claim. The theorem is stated without LayerNorm for clarity; empirical curvature measurements use the actual model including LayerNorm, so the reported constants absorb its effect.

Scope of the claim. The propositions give sufficient conditions for preserving EM conditioning and show that a shared sign-indefinite learned map removes the uniform guarantee. They do not prove necessity for all possible architectures, nor do they provide a convergence proof for the composed system.

5.4 Operations That Preserve or Destroy the Invariant

The invariant is preserved by private per-component parameters, per-coordinate monotone maps, bounded smooth kernels, and stochastic or Markov maps at least at first order. It is destroyed by dense sign-indefinite learned maps, objective mixing, hard gates, and shared lower parameters.

5.5 Classical EM Connection

Classical EM's efficiency comes from the E-step making the expected complete-data objective separate over component-private parameters. With private component parameters θ_j, the objective has the form

$$Q(\theta) = \sum_j Q_j(\theta_j),$$

and the M-step decouples into independent per-component problems. If a lower map is shared by all components, this separation disappears and the M-step becomes a coupled generalized-EM problem. Backpropagation through shared lower layers is the neural version of this classical failure.

This statement explains the structure of the foreign cross-entropy contribution, not the whole training dynamics by itself. With both local EM and supervised CE present, the update at W_1 is the sum of a private, EM-conditioned contribution and a shared, non-separable CE contribution.

Sharing manufactures the ill-conditioned term; gradient dominance decides whether that term controls the effective landscape. The limiting case in which the CE path dominates recovers the classical shared-parameter failure, while the EM-dominated case preserves conditioning despite full graph connectivity.

6 Experimental Setup

6.1 Model

The experiments use a small supervised MNIST classifier:

$$x \to \mathrm{Linear}(784, K) \to \mathrm{ReLU} \to d \to \mathrm{NLS}(d) \to \mathrm{Linear}(K, 10) \to \mathrm{LayerNorm} \to \mathrm{CE}\,.$$

Depending on the experiment, the intermediate NLS and volume-control terms are ablated. The local auxiliary objective is

$$L_{\mathrm{aux}}(d) = L_{\mathrm{LSE}}(d) + L_{\mathrm{var}}(d) + L_{\mathrm{tc}}(d).$$

6.2 Metrics

Representation health is measured by dead units, minimum variance, redundancy, and responsibility entropy. Optimization conditioning is measured by learning-rate spread, convergence-time variability, Adam versus SGD gaps, gradient norm ratios, cosine similarity between CE and EM gradients, and Hessian spectral norms with respect to the intermediate distances d.

6.3 Experimental Logic

Experiments 1–2 test at-site failures and volume-control transfer. Experiment 3 observes that the single-site conditioning signature does not survive ordinary supervised composition. Experiment 4 measures the mechanism and causally restores the property.

7 Results I: At-Site Failures Are Repaired Locally

7.1 Volume-Control Ablation

The first experiment asks whether supervision supplies enough intermediate volume control. It compares a baseline MLP, NLS alone, partial volume control, full volume control, and volume control without NLS on MNIST with $K = 25$.

Config	Dead	Min Var	Redundancy	Entropy	Accuracy
Baseline	1.40	0.000	20.5	3.08	96.09
NLS only	1.20	0.264	24.9	2.93	96.14
NLS + Var	0.00	11.74	187.0	2.09	95.75
NLS + Var + Decorr	0.00	7.84	13.9	2.46	96.34
Var + Decorr only	0.10	3.55	13.3	2.65	96.40

Table 2: Volume-control ablation summary. Accuracy is reported in percent.

Supervised gradients do not provide intermediate volume control: the baseline has dead ReLU units in every seed. NLS without volume control is insufficient. Variance alone is pathological, producing alive but highly redundant units. Full volume control repairs the site. Volume control without NLS is useful regularization, but it does not produce the same calibrated EM structure.

7.2 Capacity Sweep

The capacity sweep compares the baseline with NLS plus full volume control for $K \in \{16, 25, 36, 49, 64\}$. The baseline has dead units at every width, with a stable dead fraction around five to seven percent. Volume control eliminates dead units at every width. Redundancy grows with width, and volume control reduces the slope and tightens seed variability.

The accuracy effect is capacity- and λ-dependent. This paper therefore uses the capacity experiment as evidence for structural health rather than as a benchmark claim.

8 Results II: Conditioning Does Not Survive Composition

8.1 Optimizer Sweep

The optimizer experiment tests whether the single-layer implicit-EM optimization signature survives when the EM site is placed inside a supervised network. The model uses NLS plus full volume control, $K = 25$, $\lambda = 0.001$, SGD and Adam, and learning rates in $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$.

SGD is strongly learning-rate sensitive, spanning 83.83% to 96.23% accuracy. Adam clearly helps: Adam at learning rate 10^{-4} reaches 95.34%, and Adam at 10^{-3} reaches 96.38%. This differs from the single-layer unsupervised setting, where Adam offered little advantage.

8.2 Loss-Feature Decoupling

At high Adam learning rates, the regularization loss decreases and minimum variance explodes without meaningful accuracy gains. This partially echoes the decoder-free encoder result: the auxiliary geometry has degrees of freedom that are orthogonal to task accuracy.

8.3 Transition to Mechanism

The optimizer sweep shows that the local EM site is not enough. Experiment 4 measures why: the composed cross-entropy path dominates the local EM path, and the two gradients are nearly orthogonal.

9 Results III: The Mechanism and Causal Test

9.1 Predictions Ledger

9.2 Experiment 4 Design

The causal experiment compares four arms: joint training with $\lambda = 0.001$, joint training with $\lambda = 0.03$, joint training with $\lambda = 1$, and stop-gradient training where the supervised head reads detached intermediate distances. Feature quality is measured by a fixed-protocol linear probe trained on frozen distances.

Test	Prediction	Measured
Exp1	Supervision does not give intermediate volume control	Dead units in every baseline seed
Exp1	Partial volume control is worse than none	Redundancy increases from 20.5 to 187.0
Exp3	Optimizer signature disappears under composition	SGD spans 83.8–96.2; Adam helps
Exp4-P1	CE dominates EM at small λ	30–70x, no detectable alignment
Exp4-P2	LSE curvature stays below 1/2	Bound held; CE path 13–100x larger
Exp4-P3	Stop-gradient restores conditioning	LR spread falls from 10.9 to 1.3 points

Table 3: Prediction ledger for the supervised study.

9.3 Causal Restoration of Conditioning

Probe accuracy spread across a $1000\times$ SGD learning-rate range falls from 10.9 points in the CE-dominated joint arm to 1.3 points under stop-gradient. The $\lambda = 1$ joint arm is indistinguishable from stop-gradient, while $\lambda = 0.03$ lies between the two regimes. Supervision is not the problem; the composition corridor is.

We also track the epoch at which the test LSE reaches 95% of its final value. This is a natural stability metric in EM-dominated arms, where the LSE objective is active at the site. In the CE-dominated arm, the same metric should be read more cautiously: the local LSE is largely along for the ride rather than the dominant objective. Erratic or missing stabilization there means the EM site does not settle under CE dominance, not that the classifier failed to train.

9.4 Gradient Competition

At $\lambda = 0.001$, the CE gradient reaching the intermediate distances dominates the weighted auxiliary EM gradient by 30–$70\times$. Their cosine similarity is approximately zero. Since random vectors in $K = 25$ dimensions have cosine standard deviation about $1/\sqrt{K} \approx 0.2$, the analysis reports the full cosine distribution against this chance band. The conservative conclusion is that there is no detectable alignment: the EM signal is drowned, not opposed.

The reported dominance ratios were first measured during Adam training, whereas the conditioning sweep uses SGD. The final analysis should either add the same gradient-ratio measurement under SGD or state the optimizer mismatch explicitly. The preferred version is to measure both and report whether the ratio is stable across optimizers.

9.5 Curvature Dichotomy

The LSE Hessian with respect to d remains bounded by $1/2$ at every measured point. Under pure EM training it saturates the bound, reaching 0.4993–0.4997. The CE-path Hessian is roughly 13–$100\times$ larger, non-monotone, and consistent with W_2 spectral scaling. Distance from $1/2$ therefore acts as a diagnostic of mixture sharpness.

9.6 Lambda-Resolution Curve

The graded claim should be supported by a denser sweep over λ. The planned central figure plots conditioning spread and probe accuracy against the measured CE/EM gradient ratio, not against λ itself.

This figure also pre-empts a deflationary reading of the 88% versus 96% probe-accuracy trade-off. In the stop-gradient arm, W_1 does not receive label information, so lower probe accuracy alone would be unsurprising. The point of the denser curve is that the trade-off is continuous in the measured gradient ratio: conditioning is bought by shifting dominance, not by toggling between two unrelated training regimes.

9.7 Basin Analysis

Linear mode connectivity between EM-dominated and CE-dominated checkpoints will formalize the fine-tuning claim. Hidden units must be permutation-aligned before interpolation, for example with Hungarian matching.

9.8 Adam Arm Under Stop-Gradient

The Adam arm completes the Paper 2 optimizer signature. The prediction is that Adam provides little advantage under stop-gradient and a clear advantage in the joint CE-dominated arm.

9.9 Stochastic-Map Arm

Replacing W_2 with a non-negative row-normalized map tests the one simplex-compatible preservation class not yet covered experimentally. A positive result would provide the controlled bridge to attention's row-stochastic value path.

This arm is not load-bearing. If row normalization hurts head accuracy enough to confound the probe, the result should move to the appendix as a caveated taxonomy-cell experiment rather than appear in the abstract or headline claims.

10 Discussion

10.1 Local, Graded, Structural

The results support the main claim. Implicit EM is local: it lives at LSE/softmax sites and does not automatically propagate through learned dense maps. It is graded: the transition between CE-dominated and EM-dominated behavior follows the measured gradient ratio. It is structural: volume control fixes at-site failures, but conditioning requires private parameters or simplex-compatible transport. The structural and graded parts are complementary: the shared corridor explains why the CE-path contribution is not EM-conditioned, while gradient dominance explains when that contribution actually controls the landscape.

10.2 What Transfers and What Does Not

Collapse prevention, dead-unit prevention, redundancy control, and local mixture calibration transfer to intermediate sites. Learning-rate insensitivity, absence of Adam advantage, fixed convergence time, and parameter-independent curvature do not transfer through ordinary dense backpropagation.

10.3 Generative versus Discriminative Training

This paper is not merely another observation that generative and discriminative objectives differ. It gives a mechanism inside one architecture: a curvature dichotomy, orthogonal gradients, measured dominance, and causal restoration by stop-gradient or EM dominance.

10.4 Pretraining and Fine-Tuning

Discriminative fine-tuning can be viewed as a dominant foreign gradient that overwrites local EM structure. The basin analysis will test whether EM-dominated and CE-dominated solutions are separated by a loss barrier after permutation alignment.

10.5 Design Implications

To build deep models with EM structure, one must create local EM sites, protect them with stop-gradients or dominance-balanced objectives, or transport them through simplex-compatible maps. Ordinary learned dense maps should not be expected to preserve the property.

The value of conditioning is not primarily an output-accuracy trade. It is useful when robustness to learning-rate tuning, seed-to-seed reproducibility, stable local convergence, or elimination of dead units is worth buying, especially in settings where the EM site is trained by a layer-local objective rather than competing directly with the supervised head.

10.6 Relation to the Attention Project

The companion attention work tests the locality principle in transformers. This paper provides the controlled proof: EM conditioning does not penetrate learned linear corridors. Attention's value path is row-stochastic, so the stochastic-map arm is the controlled bridge between this study and that setting.

11 Limitations and Threats to Validity

11.1 Threats to Internal Validity

The representation-quality trade-off currently rests on a fixed-protocol linear probe. Additional probes, such as kNN or a small MLP, should test whether the ordering is probe-dependent. The curvature measurements use power iteration on fixed batches, so sensitivity to batch identity, batch size, and iteration count should be reported. The 1/2-saturation diagnostic is demonstrated at one site; claims about generality should remain cautious until measured across multiple sites.

The stop-gradient intervention changes the gradient path from CE to W_1 while holding data, labels, architecture, and CE training fixed. This makes it a clean causal knob, but the argument should be stated explicitly.

The gradient-competition measurements should be reported against the cosine distribution expected from random vectors in the same dimension, not only against zero. They should also be measured under the same optimizer used in the conditioning sweep, or the Adam-versus-SGD mismatch should be called out.

11.2 Limitations of Scope

The experiments use MNIST and small networks. The paper makes no SOTA claims and does not offer a general theory of deep-network conditioning. The formal theory gives sufficient conditions and guarantee-forfeiture statements, not a universal if-and-only-if and not a proof that composed curvature is always large. LayerNorm is included empirically but omitted from the clean proposition unless folded into the constants. No convergence proof is provided for the composed system.

12 Conclusion

Implicit EM is not a property of whole networks. It is a local structure created by LSE or softmax objectives over distances and energies. When responsibility gradients reach private parameters directly, the resulting landscape has parameter-independent curvature and exhibits the optimizer signatures observed in the single-layer model. When those gradients pass through ordinary learned dense maps, the simplex invariant is destroyed and curvature becomes learned, anisotropic, and optimizer-sensitive.

The supervised study draws the boundary from both sides. Volume control repairs local collapse and redundancy at intermediate EM sites. But conditioning survives only under isolation, dominance, or simplex-compatible structure. Knowing which properties transfer and which do not is the design rule the implicit-EM framework provides.

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